⚛️ Section: Particles and Waves

🔷 Topic: Refraction of Light

Question 12 – Refraction in Perspex

(a) Suggest how the student's measurements should be processed to find a reliable value for the refractive index

✔ 2 marks

✅ **Answer:**

> Plot a graph of $\sin i$ (angle of incidence) against $\sin r$ (angle of refraction)
> The **gradient of the graph** gives the **refractive index of Perspex**

🧠 *Why?* From Snell's Law:

$$n = \frac{\sin i}{\sin r}$$

(b) Calculate the critical angle of Perspex

✔ 3 marks

Given:

- $n_{\text{Perspex}} = 1.50$
- $n_{\text{air}} = 1.00$

Use:

$$\sin c = \frac{n_2}{n_1} = \frac{1.00}{1.50} = 0.6667$$

$$c = \sin^{-1}(0.6667) = 41.8^\circ$$

(c) Complete the ray diagram

✔ 2 marks

✅ **Answer:**

- Incident ray is at **50°** to the normal inside the Perspex
- Since $50^\circ > 41.8^\circ$, **total internal reflection** occurs
- Reflect ray **inside the Perspex**, making **equal angles to the normal**

📐 Label both angles as **50°**

🧠 Revision Tips

- **Snell's Law:**

$$n_1 \sin i = n_2 \sin r$$

- **Critical angle** is when $r = 90^\circ$ — beyond this, total internal reflection occurs
- To **graph refractive index**, plot $\sin i$ vs $\sin r$ and use the gradient